

Kian Kyars

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Python • PyTorch • TRL/vLLM • DeepSpeed • GRPO • Agent Evaluation

Experience

Judgment Labs

March 2026 - August 2026

Research Engineer

- Built a dynamic KNN+MST pipeline to cluster agent traces by failure mode (tool misuse, failed escalation) for root-cause analysis
- Built a self-improvement harness for software-engineering agents; ran 16+ hour unattended experiment loops
- Integrated a batched PII-redaction backend into the ingest pipeline, reducing cost 20×

Noemon

October 2025 - January 2026

Machine Learning Engineer

- Built a ReAct-based ARC-AGI-2 solver with tool-verified iteration and context compaction for multi-step agent workflows

fr8 Cohort 1.0

June 2025 - August 2025

- Landed [35 merged PRs](#) to [Factorio Learning Environment](#) across task-grounded agent evals, Gym infrastructure, and regression testing

Data-Driven Network and Cyber Security (DANS) Lab

September 2024 - June 2025

Undergraduate Research Assistant

- Fine-tuned Qwen2.5, Llama 3, and DeepSeek-R1 7-8B models and built local system-log anomaly detection matching GPT-4o on the evaluated task

MANGA NLP Lab

January 2025 - April 2025

Undergraduate Research Assistant

- Reimplemented the FLORA optimizer, improving 3B/7B LLM training speed up to 25% versus LoRA with AdamW

Hochschule für Technik und Wirtschaft Dresden

May 2023 - August 2023

Summer Research Intern

- DAAD RISE Intern; trained a 7-DOF Franka Emika arm in Nvidia Isaac Sim for block manipulation using TRPO and PPO

Selected Projects

Six Axes of an RL Environment

May 2026

- Designed 6 paired GRPO environments with TRL/vLLM on Qwen2.5-1.5B and ran 12 H100 experiments on Modal to isolate reward and tool-use failures

Reinforcement Learning from Verifiable Rewards (textbook)

April 2026

- Authored a 10-chapter reference on verifier design, reward hacking, training signals, and agentic RLVR

vLLM MoE Optimization (open PR)

July 2026

- Prototyped a fused Triton MoE combine reduction; measured up to 4.69× H100 microbenchmark speedup and added 33 parametrized test cases

CHIPS: From Design to Data Center

July 2026

- Published a source-checked course tracing an AI accelerator through design/EDA, fabrication, packaging, and data-center deployment

nanochat from Scratch

October 2025

- Reimplemented nanochat's tokenizer and NCCL-distributed training stack
- Additional: [OpenClaw Course](#) (250k+ views) • [Pipeline Parallelism from Scratch](#) • [Hugging Face Ultra Scale Series](#) • [UofA Run Club](#) • [UofACarpool](#)

Education

BSc Computer Science, Honours - University of Alberta

C/O 2025 Grade: 3.9/4.0